MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 2J42

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, fixed frequency, 9,345 to 9,405 MHz, 7-kw rated peak power output, integral magnet, forced-air cooled

ABSOLUTE RATINGS:

Parameter:	Ef	tk	Ib	Pi	pi	tpc	Du	rrv	T(anode)	VSWR	Alt
Unit:	V	sec	a	W	kw	µs	---	kv/µs	°C	---	ft
Maximum:	7.0	---	5.5	82.5	33	2.5	0.0025	---	120	1.5	10,000
Minimum:	---	120	3.5	---	17.5	---	---	75	--- (Note 6)	---	---

PHYSICAL CHARACTERISTICS:

See figure 1
Magnet isolation: See note 7
Output coupling: UG-40B/U
Weight: 3 pounds, approximately

TEST CONDITIONS:

Parameter:	Ef	tk	Ib	tpc	Du	rrv	VSWR
Unit:	V	sec	mAdc	µs	---	kv/µs	---
Tolerance:	---	Min	---	±10%	---	Min	Min
Test condition 1:	6.3	120	9.0	1.0	0.002	75	1.1:1
Test condition 2:	6.3 (Note 4)	120 (Note 3)	4.5	2.0	0.001	75	1.1:1 (Note 5)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1031	High-frequency vibration	---	---	No voltages applied	---	---	---	---
1047	Low-temperature operation	3	2	tk = 180 (max)	---	---	---	---
4027	Temperature coefficient	6	2	T(anode) = 40° to 70°C	ΔF/ΔT	---	0.25	MHz/°C
	Quality conformance inspection, part 1	1	---					
4289	Heater current	4	---	Ef = 6.3 V; tk = 120 (min)	If	0.43	0.60	A
4003	Pressurizing	---	---		---	45	---	psia
---	Frequency	6	1	T(anode) = 40° ±10°C; VSWR = 1.1:1	F	9,345	9,405	MHz
4250	Power output	---	1	t = 300 (max); VSWR = 1.1:1	Po	14	---	W
4306	Pulse voltage	---	1		epy	5.3	5.7	kv
4308	RF bandwidth	---	1	Ib = 7.5 and 9.0 mAdc	BW	---	2.5 tpc	MHz
		---	2	Ib = 3.5 and 4.5 mAdc	BW	---	2.5 tpc	MHz
4308	Minor lobes	---	1	Ib = 7.5 and 9.0 mAdc	Ratio	6.0	---	dB
		---	2	Ib = 3.5 and 4.5 mAdc	Ratio	6.0	---	dB
4310	Frequency pulling figure	---	1		ΔF	---	15	MHz
4315	Stability	---	1		MP	---	0.25	%
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	---	---	No voltages applied	---	---	---	---
4309	Magnetron non-oscillating impedance:							
	Standing-wave ratio	2	---		σ'	8:1	---	---
	Position of maximum	2	---		Distance	13.5	22.5	mm
4315	Stability	8	2		MP	---	0.25	%
	Quality conformance inspection, part 3							
---	Life test	---	2	Group D	t	250	---	hrs
---	Life-test end points:	---	---					
---	Frequency	6	1	T(anode) = 40° ±10°C; VSWR = 1.1:1	F	9,345	9,405	MHz
4250	Power output	---	1	t = 300 (max); VSWR = 1.1:1	Po	11	---	W
4308	RF bandwidth	---	1		BW	---	3.0 tpc	MHz
4315	Stability	8	2		MP	---	1.0	%

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. The first standing wave minimum shall be located on the waveguide side of reference plane "A", and the measurement taken between these points. When this number is subtracted from one-half waveguide wavelength, the remainder (length from point "A" to the minimum toward the tube) should fall within the limits specified. The hot-test frequency of the magnetron shall be used in this test.

3. The cathode heating time shall be a minimum of 120 seconds at an ambient temperature of greater than 0°C, and a minimum of 180 seconds at temperatures between 0° and -55°C.

4. For average pulse power inputs in excess of 25 watts, the heater voltage shall be reduced within 3 seconds after applying high voltage according to the following formula:

$$Ef = 6.3 \quad ^{(1-\frac{Pi}{180})} \quad \text{volts}$$

where:

Pi = Average pulse power input in watts.

At no time shall the heater current be allowed to exceed a surge value of 3 amperes.

5. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.

6. The anode temperature shall be measured at the point indicated on figure 1.

7. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 2 inches (50.8 mm) of the tube.

8. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percent of the number of input pulses applied during the period of observation. Pulses are considered to be "missing" if, due to any cause, their rf energy is less than 70 percent of the normal energy level in the frequency range of 9,345 to 9,405 MHz. The number of missing pulses shall not exceed the amount specified, expressed as a percent of the number of applied pulses, during any 5-minute interval of a 15-minute test period.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

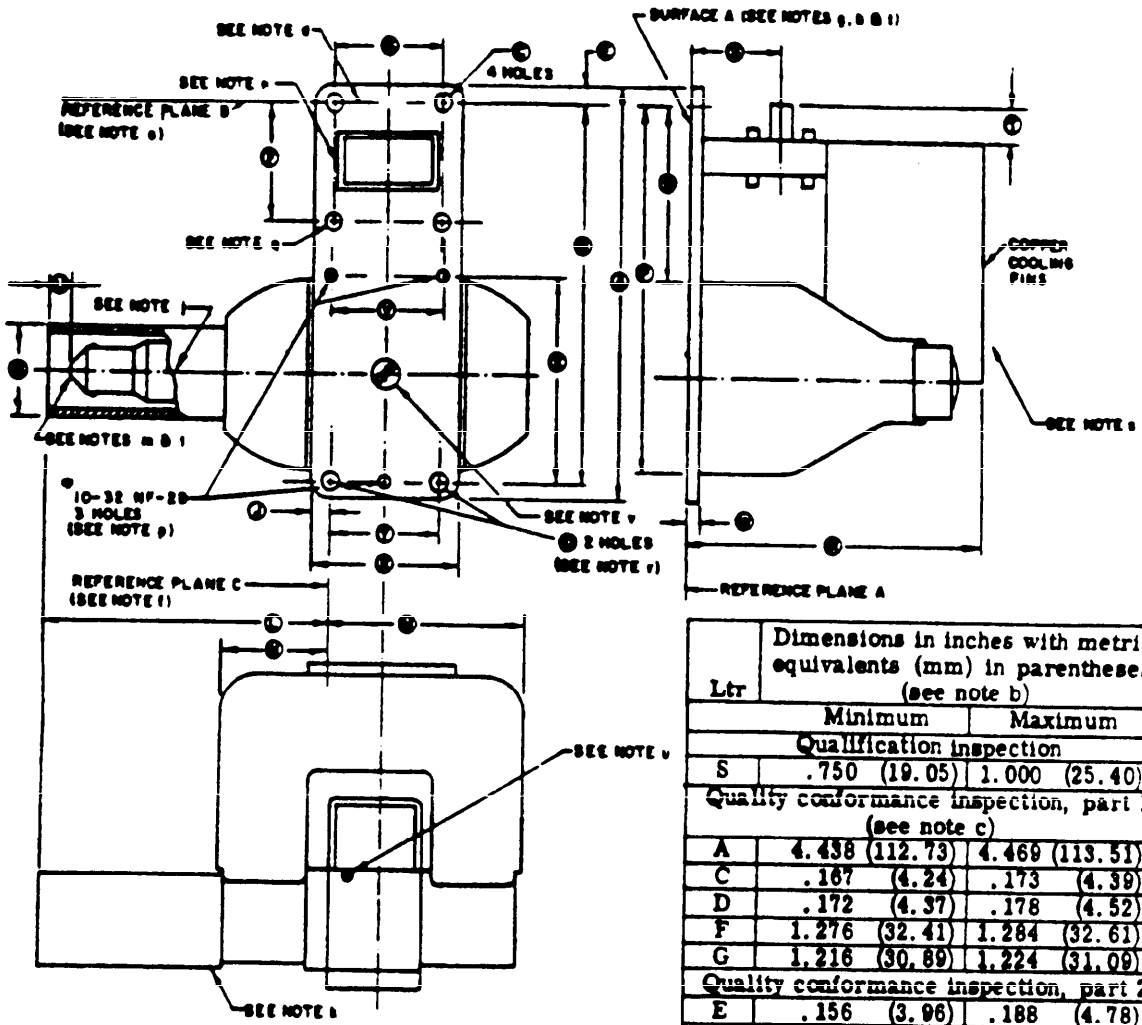
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
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(Project 5960-2970)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note b)	
	Minimum	Maximum
Qualification inspection		
S	.750 (19.05)	1.000 (25.40)
Quality conformance inspection, part 1 (see note c)		
A	4.438 (112.73)	4.469 (113.51)
C	.167 (4.24)	.173 (4.39)
D	.172 (4.37)	.178 (4.52)
F	1.276 (32.41)	1.284 (32.61)
G	1.216 (30.89)	1.224 (31.09)
Quality conformance inspection, part 2		
E	.156 (3.96)	.188 (4.78)
H		1.000 (25.40)
J	.188 (4.78)	.219 (5.56)
K	1.609 (40.88)	1.641 (41.68)
L	2.688 (68.28)	3.188 (80.98)
M		2.188 (55.58)
N		1.166 (29.16)
P		4.000 (101.60)
Q	1.938 (49.23)	
R		3.313 (84.15)
T		.375 (9.53)
U		.250 (6.35)
Nominal dimensions		
B	4.103 (104.22)	
V	1.250 (31.75)	
W	.125 (3.18)	
X	2.393 (60.78)	
Y	1.220 (30.99)	

FIGURE 1. Outline drawing of electron tube type 2J42.

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NOTES.
a. All dimensions are in inches.
b. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
c. The AQL for the combined mechanical defectives in quality conformance inspection, part 1, shall be 1.0 percent.
d. This area is gasketed for pressurizing waveguide output, as with coupler UG-40B/U, in accordance with MIL-F-3922/59, and shall be tested in accordance with method 4003.
e. Reference plane B passes through centers of two top holes of mounting plate as shown and is perpendicular to reference plane A.
f. Reference plane C passes through the upper left hole on mounting plate as shown and is mutually perpendicular to reference planes A and B.
g. With surface A resting on a flat-surface plate, a feeler gage .020 thick and 1/8 wide shall not enter more than 1/8 at any point.
h. Surface A and interior surfaces of waveguide shall be gold plated 10 MSI or silver plated 30 MSI. Plating is not required over parts made of monel, or equivalent, corrosion resistant material.
j. Axis of filament lead protector shall lie within 5° of a normal to reference plane C.
k. The filament lead protector shall not be used to support any cap fitting. Protector is a detachable sleeve of nonconducting material.
m. Clearance between inside surface of filament lead protector and .375 diameter cylindrical surface of standard single contact miniature bayonet lamp base shall be not less than .125.
n. Position of waveguide hole not specified since tubes are tested and used with coupler UG-40B/U. (See note d)
p. Holes shall be located within .015 of indicated centers.
q. Center of this hole shall lie within .004 of reference plane C.
r. Holes shall lie within .005 of indicated centers. A cylinder .330 diameter, centered in holes shown, shall clear the side of the magnet.
s. Recommended direction of air blast.
t. All metal surfaces except surface A and the bayonet lamp base shall be painted with heat-resistant paint.
u. Anode temperature shall be measured at this point.
v. Soft solder shall be used. As an alternate, tip of screw may be soldered.

FIGURE 1. Outline drawing of electron tube type 2J42 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N 0102-014-1801 C-2531

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